
Examiners' commentaries 2025 (October)

ST2134 Advanced statistics: statistical inference

Important note

This commentary reflects the examination and assessment arrangements for this course in the academic year 2024–25. The format and structure of the examination may change in future years, and any such changes will be communicated.

Information about the course and the Essential reading references

Unless otherwise stated, all cross-references will be to the latest version of the course (2021). You should always attempt to use the most recent edition of any Essential reading textbook, even if the commentary and/or online reading list refer to an earlier edition. If different editions of Essential reading are listed, please check the VLE/Digital Campus for reading supplements – if none are available, please use the contents list and index of the new edition to find the relevant section. You will also find the Learning Outcomes for the course on the VLE/Digital Campus.

General remarks

What the examiners are looking for

Candidates should pay close attention to the essential topics of sufficient statistics, point estimation, interval estimation and hypothesis testing. Being able to state and prove key theorems and lemmas simply requires close study of these in the subject guide. The point estimation techniques of method of moments estimation and maximum likelihood estimation are standard (and are even introduced in **ST104B Statistics 2** and **ST1215 Introduction to mathematical statistics**), with questions typically varying in terms of the probability distribution from which the random sample is drawn (with many common distributions covered in **ST104B Statistics 2**, **ST1215 Introduction to mathematical statistics** and **ST2133 Advanced statistics: distribution theory**). Pivotal functions feature heavily in Chapter 4 (Interval estimation), and likelihood ratio tests similarly feature heavily in Chapter 5 (Hypothesis testing).

An examination paper attempts to cover as much of the syllabus as possible, but inevitably some parts are left out. As the included parts will vary from year to year it is not enough to practise only on the past papers. Candidates should practise on all the examples, activities and sample examination questions in the subject guide in order to understand the relevant theory and adequately prepare for the examination. Afterwards you may try one or two past papers for further practice.

The examination paper for this course involves four questions and each of the questions contains several parts which contribute towards the final grade with the different numbers of marks reflecting their difficulty. The difficulty of each part is usually assessed after taking into account the level of thinking related to statistical inference and the extent of algebraic computations required. Take 5–10 minutes to read the whole paper and try to identify the difficulties of each question in comparison to your ability. You will find that some questions could be much easier for you than others.

If you are unable to solve an exercise due to algebraic computations or lack of knowledge of the material of **ST2133 Advanced statistics: distribution theory**, do not abandon an exercise. Write clearly the procedure which should be followed for the relevant statistical inference task(s). Remember that the primary aspect you are being examined on is the ability to *understand* the statistical methodology.

A good knowledge of basic calculus, such as the ability to compute integrals, sums and derivatives, will give you invaluable help for completing the examination paper. Also, the main concepts of **ST2133 Advanced statistics: distribution theory** (random variables and their distribution, expectation, independence, conditional expectation etc.) should be well-understood before attempting this half course. If you are not comfortable with any of these, go back and revise. Memorising basic properties such as the moments and probability mass/density functions of some standard distributions (such as the binomial, Poisson, uniform, normal and gamma) is also a good idea.

When you are trying to prove an equality or inequality be careful of what is on the left-hand and right-hand sides. Justify the steps in the calculations as much as possible. If you cannot get a question right, do not just write in the correct answer without any further explanation. If you just write on the script what you think went wrong you will get more marks. Also, try to avoid writing more than you are asked to write. For example, when asked to *state* a theorem, do not give the proof as well as this simply wastes time.

Be clear in your answers. When giving the probability mass/density function of a random variable you should also state its sample space (also known as its support). When you are deriving an estimator or a sufficient statistic also state the parameter to which it corresponds. Sometimes it is a good idea to underline your final result (if applicable).

Make sure that you are able to provide the definitions of basic concepts such as estimators vs. estimates, confidence intervals and hypothesis tests. Remember that hypotheses are statements about population parameters which are estimated using functions of the data (that is, *statistics*). In the relevant calculations you should always be able to distinguish a statistic from a parameter and be clear on what they represent, respectively.

Practise operations involving products and sums. A substantial number of candidates fail to write them down correctly. Quite often they omit or write the indices incorrectly which inevitably leads to errors. A typical example of this is the following:

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a population with probability density function $f_{X_i}(x_i; \theta)$, often written for simplicity as $f_X(x; \theta)$. An observed sample is $\mathbf{x} = \{x_1, x_2, \dots, x_n\}$. The fact that this sample is a random sample indicates that we can write the joint probability density function of $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ as:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = \prod_{i=1}^n f_{X_i}(x_i; \theta).$$

Make sure that you write the above and *not* the following:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = \prod_{i=1}^n f_X(x; \theta)$$

which can lead to the *incorrect* expression:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = (f_X(x; \theta))^n.$$

Note that if you were asked to provide the log-likelihood function $l_{\mathbf{X}}(\theta; \mathbf{x})$ the calculations would be formed of *sums* rather than *products* (refresh your memory on the properties of logarithmic functions). The correct expression would be:

$$l_{\mathbf{X}}(\theta; \mathbf{x}) = \log(f_{\mathbf{X}}(\theta; \mathbf{x})) = \log\left(\prod_{i=1}^n f_{X_i}(x_i; \theta)\right) = \sum_{i=1}^n \log f_{X_i}(x_i; \theta).$$

Examination revision strategy

Many candidates are disappointed to find that their examination performance is poorer than they expected. This can be due to a number of different reasons and the *Examiners' commentaries* suggest ways of addressing common problems and improving your performance. We want to draw your attention to one particular failing – ‘question spotting’, that is, confining your examination preparation to a few question topics which have come up in past papers for the course. This can have very serious consequences.

We recognise that candidates may not cover all topics in the syllabus in the same depth, but you need to be aware that examiners are free to set questions on **any aspect** of the syllabus. This means that you need to study enough of the syllabus to enable you to answer the required number of examination questions.

The syllabus can be found in the Course information sheet in the section of the VLE dedicated to this course. You should read the syllabus very carefully and ensure that you cover sufficient material in preparation for the examination. Examiners will vary the topics and questions from year to year and may well set questions that have not appeared in past papers – every topic on the syllabus is a legitimate examination target. So although past papers can be helpful in revision, you cannot assume that topics or specific questions that have come up in past examinations will occur again.

If you rely on a question spotting strategy, it is likely you will find yourself in difficulties when you sit the examination paper. We strongly advise you not to adopt this strategy.

Comments on specific questions

Candidates should answer all **FOUR** questions: **Question 1** of Section A (40 marks) and **all THREE** questions from Section B (60 marks in total). **Candidates are strongly advised to divide their time accordingly.**

Section A

Answer Question 1 from this section.

Question 1

- (a) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a continuous probability distribution parameterised by θ with log-likelihood function $l_{\mathbf{X}}(\theta; \mathbf{x})$. Show that Fisher information about θ in the random sample $\mathbf{X}$ can also be expressed as:

$$\mathcal{I}_{\mathbf{X}}(\theta) = -\mathbf{E} \left(\frac{\partial^2}{\partial \theta^2} l_{\mathbf{X}}(\theta; \mathbf{X}) \right) = -\mathbf{E} \left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) \right).$$

(10 marks)

Reading for this question

Section 2.10.2 of the course presents the relevant content on the score function and Fisher information.

Approaching the question

We proceed by differentiating $E(s_{\mathbf{X}}(\theta; \mathbf{X}))$ with respect to θ . We have:

$$\begin{aligned}
 0 &= \frac{d}{d\theta} E(s_{\mathbf{X}}(\theta; \mathbf{X})) \\
 &= \frac{d}{d\theta} \int_{\mathbb{R}^n} s_{\mathbf{X}}(\theta; \mathbf{X}) f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x} \\
 &= \int_{\mathbb{R}^n} \frac{d}{d\theta} (s_{\mathbf{X}}(\theta; \mathbf{X}) f_{\mathbf{X}}(\mathbf{x}; \theta)) d\mathbf{x} \\
 &= \int_{\mathbb{R}^n} \left(\left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) \right) f_{\mathbf{X}}(\mathbf{x}; \theta) + s_{\mathbf{X}}(\theta; \mathbf{X}) \frac{\partial}{\partial \theta} f_{\mathbf{X}}(\mathbf{x}; \theta) \right) d\mathbf{x} \\
 &= \int_{\mathbb{R}^n} \left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) + (s_{\mathbf{X}}(\theta; \mathbf{X}))^2 \right) f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x} \\
 &= E \left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) + (s_{\mathbf{X}}(\theta; \mathbf{X}))^2 \right) \\
 &= E \left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) \right) + E((s_{\mathbf{X}}(\theta; \mathbf{X}))^2).
 \end{aligned}$$

since:

$$\frac{\partial}{\partial \theta} f_{\mathbf{X}}(\mathbf{x}; \theta) = s_{\mathbf{X}}(\theta; \mathbf{X}) f_{\mathbf{X}}(\mathbf{x}; \theta).$$

Hence:

$$E((s_{\mathbf{X}}(\theta; \mathbf{X}))^2) = -E \left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) \right)$$

and so:

$$\mathcal{I}_{\mathbf{X}}(\theta) = E((s_{\mathbf{X}}(\theta; \mathbf{X}))^2) = -E \left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) \right).$$

Common errors: Failing to justify interchanging differentiation and integration; not stating that $E(s_{\mathbf{X}}(\theta; \mathbf{X})) = 0$; omitting the product rule step; unclear notation between $d/d\theta$ and $\partial/\partial\theta$; missing the final step showing $\partial s_{\mathbf{X}}(\theta; \mathbf{X})/\partial\theta = \partial^2 l_{\mathbf{X}}(\theta; \mathbf{x})/\partial\theta^2$.

Excellent answers explicitly invoked regularity conditions; clearly stated $E(s_{\mathbf{X}}(\theta; \mathbf{X})) = 0$ before differentiating; showed all product rule steps; recognised $s_{\mathbf{X}}(\theta; \mathbf{X}) = \partial l_{\mathbf{X}}(\theta; \mathbf{x})/\partial\theta$ implies $\partial s_{\mathbf{X}}(\theta; \mathbf{X})/\partial\theta = \partial^2 l_{\mathbf{X}}(\theta; \mathbf{x})/\partial\theta^2$; maintained precise notation throughout; connected both required expressions to the Fisher information definition.

(b) i. Provide the definition of a sufficient statistic.

(4 marks)

ii. State the factorisation theorem for the continuous case.

(6 marks)

Reading for this question

Sections 2.8.1 and 2.8.3 of the course introduce sufficient statistics and the factorisation theorem, respectively.

Approaching the question

- Let $\mathbf{X}$ be a random sample parameterised by θ . A statistic $\mathbf{U} = \mathbf{T}(\mathbf{X})$ is a sufficient statistic for θ if the conditional distribution of $\mathbf{X}$ given $\mathbf{U}$ does not depend on θ .
- Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample with joint density function $f_{\mathbf{X}}(\mathbf{x}; \theta)$. The statistic $\mathbf{U} = \mathbf{T}(\mathbf{X})$ is sufficient for the vector of parameters θ if and only if we can find functions g and h such that:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = g(\mathbf{T}(\mathbf{x}), \theta) h(\mathbf{x})$$

for all $\mathbf{x} \in \mathbb{R}^n$ and $\theta \in \Theta$. Note $\mathbf{T}(\mathbf{X})$ and θ can be scalars, denoted $T(\mathbf{X})$ and θ , respectively.

Common errors (i): Vague statements like ‘contains all information’ without specifying the conditional distribution; omitting ‘does not depend on θ ’; confusing sufficiency with parameter estimation. In part (ii): Missing ‘if and only if’; not clarifying h is θ -free; failing to state g depends on $\mathbf{x}$ only through $T(\mathbf{x})$; omitting domain specifications.

Excellent answers had precise conditional distribution statement; explicit θ -independence; correct biconditional phrasing; clear functional dependencies (g through $T(\mathbf{x})$ and θ ; h through $\mathbf{x}$ only); proper domain statements; clear notation throughout both definitions.

- (c) A random sample consists of a single observation of X which follows a binomial distribution, specifically $\text{Bin}(8, \pi)$. Suppose we are interested in testing:

$$H_0 : \pi = 0.50 \quad \text{vs.} \quad H_1 : \pi = 0.75.$$

A statistical test rejects H_0 if the observed value x is 8. Calculate the *size* and the *power* of the test.

(8 marks)

Reading for this question

Section 5.4.3 of the course presents relevant definitions of the size and power of statistical tests.

Approaching the question

To find the size of the test we need to find the probability given by:

$$P(\text{reject } H_0 \mid \pi = 0.50)$$

and for the power we need to calculate the probability given by:

$$P(\text{reject } H_0 \mid \pi = 0.75).$$

Since we reject H_0 if $x = 8$ and $X \sim \text{Bin}(8, \pi)$, the size of the test is:

$$P(\text{reject } H_0 \mid \pi = 0.50) = P(X = 8 \mid \pi = 0.50) = \binom{8}{8} (0.50)^8 = \frac{1}{256} = 0.0039$$

and the power of the test is:

$$P(\text{reject } H_0 \mid \pi = 0.75) = P(X = 8 \mid \pi = 0.75) = \binom{8}{8} (0.75)^8 = 0.1001.$$

Common errors: Confusing size and power definitions; calculating $P(\text{Type II error})$ instead of power; using the wrong parameter value (for example, $\pi = 0.75$ for size); arithmetic mistakes in binomial calculations; not clearly identifying which hypothesis applies to each probability.

Excellent answers explicitly stated definitions (size = $P(\text{reject } H_0 \mid H_0 \text{ true})$; power = $P(\text{reject } H_0 \mid H_1 \text{ true})$); clearly identified parameter values before calculating; showed the complete binomial formula; provided both exact fractions and decimal approximations; demonstrated understanding that size measures the Type I error rate.

- (d) Let $\mathbf{X} = \{X_1, X_2\}$ be a random sample of size 2 from a Bernoulli distribution with parameter π . Consider the two statistics:

$$S(\mathbf{X}) = X_1 + X_2 \quad \text{and} \quad T(\mathbf{X}) = X_1 X_2.$$

Prove *without using the factorisation theorem* that $S(\mathbf{X})$ is a sufficient statistic for π , but $T(\mathbf{X})$ is not.

Hint: Consider the possible values that each statistic can take, and determine:

$$P(X_1 = x_1, X_2 = x_2 \mid S = s) \quad \text{and} \quad P(X_1 = x_1, X_2 = x_2 \mid T = t).$$

(12 marks)

Reading for this question

Section 2.8 of the course covers sufficiency.

Approaching the question

Consider first $S(\mathbf{X})$, which can take values $s = 0, 1, 2$ and is distributed as $\text{Bin}(2, \pi)$. Hence for any $x_1, x_2 = 0, 1$ we have:

$$\begin{aligned} P(X_1 = x_1, X_2 = x_2 | S = s) &= \frac{P(X_1 = x_1, X_2 = x_2, S = s)}{P(S = s)} \\ &= \frac{P(X_1 = x_1, X_2 = x_2)}{P(S = s)} \\ &= \frac{P(X_1 = x_1) P(X_2 = x_2)}{P(S = s)} \\ &= \frac{\pi^{x_1} (1 - \pi)^{1-x_1} \pi^{x_2} (1 - \pi)^{1-x_2}}{\binom{2}{s} \pi^s (1 - \pi)^{2-s}} \\ &= \frac{\pi^{x_1+x_2} (1 - \pi)^{2-(x_1+x_2)}}{\binom{2}{s} \pi^s (1 - \pi)^{2-s}} \end{aligned}$$

which does not depend on π since $s = x_1 + x_2$, hence $S(\mathbf{X})$ is sufficient for π .

Turning to $T(\mathbf{X})$ it can take values $t = 0, 1$. Consider, for example:

$$P(X_1 = x_1, X_2 = x_2 | T = 0) = \frac{P(X_1 = x_1) P(X_2 = x_2)}{P(T = 0)} = \frac{\pi^{x_1+x_2} (1 - \pi)^{2-(x_1+x_2)}}{2\pi(1 - \pi) + (1 - \pi)^2}$$

which does depend on π , therefore $T(\mathbf{X})$ is not sufficient for π .

Common errors: Using the factorisation theorem despite prohibition; not checking all possible statistic values; failing to recognise that $P(X_1 = x_1, X_2 = x_2, S = s)$ equals $P(X_1 = x_1, X_2 = x_2)$ when $s = x_1 + x_2$; incomplete simplification obscuring π -cancellation for S ; not providing specific numerical examples for T ; arithmetic errors in conditional probabilities.

Excellent answers systematically enumerated all statistic values; explicitly showed $P(S = s)$ follows $\text{Bin}(2, \pi)$; demonstrated complete algebraic cancellation of π terms for S ; provided multiple cases for T showing π -dependence; clearly invoked independence throughout; accurate conditional probability calculations; definitive conclusions linking back to sufficiency definition.

Section B

Answer all three questions from this section.

Question 2

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a distribution with probability density function:

$$f_X(x; \theta) = \begin{cases} \frac{x}{2\theta^2} & 0 \leq x \leq 2\theta \\ 0 & \text{otherwise} \end{cases}$$

where $\theta > 0$ is an unknown parameter.

(a) Determine a method of moments estimator of θ .

(4 marks)

- (b) Derive the maximum likelihood estimator of θ , and determine the sampling distribution of the sample maximum.

(8 marks)

- (c) Compare the method of moments and maximum likelihood estimators derived in parts (a) and (b), respectively, in terms of bias. Can the maximum likelihood estimator be improved upon in terms of bias? If so, demonstrate how.

(8 marks)

Reading for this question

Sections 3.6 and 3.9 of the course cover method of moments estimation and maximum likelihood estimation, respectively.

Approaching the question

- (a) The first population moment is:

$$E(X) = \int_0^{2\theta} \frac{x^2}{2\theta^2} dx = \frac{1}{2\theta^2} \left[\frac{x^3}{3} \right]_0^{2\theta} = \frac{8\theta^3}{6\theta^2} = \frac{4\theta}{3}.$$

We estimate $E(X)$ with the sample mean, hence:

$$\widehat{E(X)} = \bar{X} \implies \frac{4\hat{\theta}}{3} = \bar{X} \implies \hat{\theta}_{MM} = \frac{3\bar{X}}{4}.$$

- (b) Due to independence, the likelihood function is:

$$L_{\mathbf{X}}(\theta; \mathbf{x}) = \prod_{i=1}^n \frac{x_i}{2\theta^2} I(x_i \leq 2\theta) \propto \theta^{-2n} I(x_{(n)} \leq 2\theta)$$

where $x_{(n)}$ denotes the sample maximum.

The likelihood function takes positive non-zero values for $\theta \geq x_{(n)}/2$ such that:

$$L_{\mathbf{X}}(\theta; \mathbf{x}) = \begin{cases} (2\theta)^{-n} & \text{for } \theta \geq x_{(n)}/2 \\ 0 & \text{otherwise.} \end{cases}$$

The likelihood function is decreasing in θ so is maximised at the smallest possible value of θ , which is $x_{(n)}/2$. Hence the maximum likelihood estimator of θ is

$$\hat{\theta}_{ML} = X_{(n)}/2.$$

To find the distribution of $X_{(n)}$, we first find the cdf of X , given by:

$$F_X(x; \theta) = \int_0^x \frac{t}{2\theta^2} dt = \frac{1}{2\theta^2} \left[\frac{t^2}{2} \right]_0^x = \frac{x^2}{4\theta^2}$$

for $0 \leq x \leq 2\theta$. The general formula for the pdf of the sample maximum is:

$$f_{X_{(n)}}(x; \theta) = n(F_X(x; \theta))^{n-1} f_X(x; \theta).$$

Hence:

$$f_{X_{(n)}}(x; \theta) = n \left(\frac{x^2}{4\theta^2} \right)^{n-1} \frac{x}{2\theta^2} = \frac{nx^{2n-1}}{2^{2n-1}\theta^{2n}}$$

for $0 \leq x \leq 2\theta$, and 0 otherwise.

(c) The method of moments estimator is unbiased since:

$$E(\hat{\theta}_{MM}) = E\left(\frac{3\bar{X}}{4}\right) = \frac{3}{4} E(\bar{X}) = \frac{3}{4} E(X) = \frac{3}{4} \frac{4\theta}{3} = \theta.$$

The expected value of $\hat{\theta}_{ML} = X_{(n)}/2$ is:

$$\begin{aligned} E\left(\frac{X_{(n)}}{2}\right) &= \int_0^{2\theta} \frac{1}{2} \frac{nx^{2n}}{2^{2n-1}\theta^{2n}} dx = \frac{n}{2^{2n}\theta^{2n}} \left[\frac{x^{2n+1}}{2n+1}\right]_0^{2\theta} \\ &= \frac{n}{2^{2n}\theta^{2n}} \frac{2^{2n+1}\theta^{2n+1}}{2n+1} \\ &= \frac{2n}{2n+1} \theta \\ &\neq \theta. \end{aligned}$$

Hence the maximum likelihood estimator is biased and will not be better than the method of moments estimator in terms of bias.

However, if we set:

$$\tilde{\theta} = \frac{2n+1}{2n} \hat{\theta}_{ML} = \frac{2n+1}{4n} X_{(n)} \implies E(\tilde{\theta}) = \theta$$

we would get an unbiased estimator.

Common errors (a): Incorrect integration limits or integrand; arithmetic mistakes yielding wrong $E(X)$. In (b): Not recognising the support constraint $\theta \geq x_{(n)}/2$; attempting differentiation instead of boundary analysis; errors deriving order statistic pdf. In (c): Computational errors in $E(X_{(n)})$; not recognising the maximum likelihood estimator bias; incorrect bias adjustment factor; failing to show that the $(2n+1)/(2n)$ correction yields unbiasedness.

Excellent answers showed clear integration throughout; clear likelihood monotonicity argument; complete order statistic derivation; both bias calculations shown; explicit bias correction with verification; understanding that boundary maximum likelihood estimators often require adjustment.

Question 3

- (a) Let $\mathbf{X} = \{X_1, X_2, \dots, X_{10}\}$ be a random sample from a distribution with mean μ . Suppose $\sum_{i=1}^{10} X_i$ is a sufficient statistic for μ . Based on the initial estimator $\hat{\mu} = (X_1 + X_2)/2$, use the Rao–Blackwell theorem to find an improved estimator of μ .

(6 marks)

Reading for this question

Section 3.5 of the course presents the Rao–Blackwell theorem.

Approaching the question

Since $X_1, X_2, \dots, X_{10}$ are identically distributed:

$$E\left(X_i \mid \sum_{i=1}^{10} X_i\right) = E\left(X_1 \mid \sum_{i=1}^{10} X_i\right).$$

Using the Rao–Blackwell theorem, if we condition on the sufficient statistic $\sum_{i=1}^{10} X_i$, we get the following improved estimator:

$$\begin{aligned}
 E\left(\hat{\mu} \mid \sum_{i=1}^{10} X_i\right) &= E\left(\frac{X_1 + X_2}{2} \mid \sum_{i=1}^{10} X_i\right) \\
 &= \frac{1}{2} E\left(X_1 \mid \sum_{i=1}^{10} X_i\right) + \frac{1}{2} E\left(X_2 \mid \sum_{i=1}^{10} X_i\right) \\
 &= E\left(X_1 \mid \sum_{i=1}^{10} X_i\right) \\
 &= E\left(X_i \mid \sum_{i=1}^{10} X_i\right) \\
 &= \frac{1}{10} \sum_{i=1}^{10} E\left(X_i \mid \sum_{i=1}^{10} X_i\right) \\
 &= \frac{1}{10} E\left(\sum_{i=1}^{10} X_i \mid \sum_{i=1}^{10} X_i\right) \\
 &= \frac{1}{10} \sum_{i=1}^{10} X_i \\
 &= \bar{X}.
 \end{aligned}$$

So the improved estimator is $\bar{X}$.

Common errors: Not explicitly invoking symmetry of identically distributed variables; skipping the important step showing all $E(X_i \mid \sum_i X_i)$ are equal; computational errors in conditional expectation algebra; not clearly stating the Rao–Blackwell theorem application; missing the insight that $\sum_i E(X_i \mid S) = E(\sum_i X_i \mid S)$.

Excellent answers stated the explicit symmetry statement upfront; clear Rao–Blackwell invocation; systematic algebraic progression showing linearity of conditional expectation; recognition that factoring 1/10 enables simplification; concluding that sample mean is the improved estimator; demonstrating understanding that conditioning on sufficient statistics reduces variance while preserving unbiasedness.

- (b) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from the distribution with probability density function:

$$f_X(x; \mu) = \left(\frac{1}{2\pi x^3}\right)^{1/2} \exp\left(\frac{1}{\mu} - \frac{x}{2\mu^2} - \frac{1}{2x}\right)$$

for $x > 0$, and 0 otherwise, where $\mu > 0$ is an unknown parameter.

- i. Derive the likelihood function and provide a one-dimensional sufficient statistic for μ .

(4 marks)

- ii. Derive the maximum likelihood estimator of μ .

(4 marks)

- iii. Find an approximate pivotal function for μ and construct an asymptotic 100(1 - α)% confidence interval for μ with $\alpha \in (0, 1)$.

(6 marks)

Reading for this question

Section 3.9 of the course covers maximum likelihood estimation, and Section 4.6 explains how to find interval estimators from pivotal functions.

Approaching the question

- i. Due to independence, the likelihood function is:

$$\begin{aligned} L_{\mathbf{X}}(\mu; \mathbf{x}) &= \prod_{i=1}^n f_X(x_i; \mu) = \prod_{i=1}^n \left(\frac{1}{2\pi x_i^3} \right)^{1/2} \exp \left(\frac{1}{\mu} - \frac{x_i}{2\mu^2} - \frac{1}{2x_i} \right) \\ &= \left(2\pi \prod_{i=1}^n x_i^3 \right)^{-n/2} \exp \left(\frac{n}{\mu} - \frac{\sum_{i=1}^n x_i}{2\mu^2} - \frac{1}{2} \sum_{i=1}^n \frac{1}{x_i} \right) \\ &= \exp \left(\frac{n}{\mu} - \frac{\sum_{i=1}^n x_i}{2\mu^2} \right) \exp \left(-\frac{1}{2} \sum_{i=1}^n \frac{1}{x_i} \right) \left(2\pi \prod_{i=1}^n x_i^3 \right)^{-n/2}. \end{aligned}$$

We can write $f_{\mathbf{X}}(\mathbf{x}; \mu) = g(\sum_i x_i, \mu) h(\mathbf{x})$ by setting:

$$g \left(\sum_{i=1}^n x_i, \mu \right) = \exp \left(\frac{n}{\mu} - \frac{\sum_{i=1}^n x_i}{2\mu^2} \right)$$

and:

$$h(\mathbf{x}) = \exp \left(-\frac{1}{2} \sum_{i=1}^n \frac{1}{x_i} \right) \left(2\pi \prod_{i=1}^n x_i^3 \right)^{-n/2}.$$

Hence $\sum_{i=1}^n X_i$ is a one-dimensional sufficient statistic for μ .

- ii. Note that the likelihood function is proportional to:

$$L_{\mathbf{X}}(\mu; \mathbf{x}) \propto \exp \left(\frac{n}{\mu} - \frac{\sum_{i=1}^n x_i}{2\mu^2} \right).$$

The log-likelihood is therefore:

$$l_{\mathbf{X}}(\mu; \mathbf{x}) = \log L_{\mathbf{X}}(\mu; \mathbf{x}) \propto \frac{n}{\mu} - \frac{\sum_{i=1}^n x_i}{2\mu^2}.$$

The score function is:

$$s_{\mathbf{X}}(\mu; \mathbf{x}) = \frac{\partial}{\partial \mu} l_{\mathbf{X}}(\mu; \mathbf{x}) = -\frac{n}{\mu^2} + \frac{\sum_{i=1}^n x_i}{\mu^3}.$$

Setting $s_{\mathbf{X}}(\mu; \mathbf{x}) = 0$ provides $\hat{\mu} = \sum_i X_i / n = \bar{X}$ as a candidate maximum likelihood estimator. Since:

$$\left. \frac{\partial^2}{\partial \mu^2} l_{\mathbf{X}}(\mu; \mathbf{x}) \right|_{\mu=\bar{x}} = \frac{2n}{\mu^3} - \frac{3n\bar{x}}{\mu^4} \Big|_{\mu=\bar{x}} = \frac{2n}{\bar{x}^3} - \frac{3n\bar{x}}{\bar{x}^4} = -\frac{n}{\bar{x}^3} < 0$$

we conclude that this is indeed the maximum likelihood estimator.

iii. We know that for large sample sizes:

$$\hat{\mu} \sim N\left(\mu, \frac{1}{I_{\mathbf{X}}(\mu)}\right)$$

approximately. In an extra level of approximation, we may use:

$$\hat{\mu} \sim N\left(\mu, \frac{1}{H(\mu; \mathbf{x})}\right)$$

approximately, where:

$$H(\mu; \mathbf{x}) = - \left. \frac{\partial^2}{\partial \mu^2} l_{\mathbf{X}}(\mu; \mathbf{x}) \right|_{\mu=\bar{x}} = \frac{n}{\bar{x}^3}.$$

Let $S_{\mu} = \bar{x}^3/n$. We can then write:

$$\frac{\hat{\mu} - \mu}{\sqrt{S_{\mu}}} \sim N(0, 1)$$

approximately. Using the pivotal function technique for constructing confidence intervals, we get the following asymptotic $100(1 - \alpha)\%$ confidence interval:

$$\left(\hat{\mu} - z_{\alpha/2} \sqrt{S_{\mu}}, \hat{\mu} + z_{\alpha/2} \sqrt{S_{\mu}}\right)$$

where $z_{\alpha/2}$ is the top 100α th percentile of $N(0, 1)$.

Common errors (i): Algebraic mistakes simplifying exponential products; incomplete factorisation; not explicitly identifying the g and h functions; missing that $\sum_i X_i$ is sufficient. In part (ii): Differentiation errors in the score function; not verifying the second derivative condition; arithmetic mistakes showing the second derivative is negative. In part (iii): Confusing Fisher information with observed information H ; incorrect H calculation; wrong standardisation in pivotal quantity; using z_{α} instead of $z_{\alpha/2}$.

Excellent answers had clean exponential algebra with all steps; explicit factorisation theorem application; systematic score derivation with verification; clear invocation of asymptotic normality; proper observed information $H = n/\bar{x}^3$; correct pivotal function $(\hat{\mu} - \mu)/\sqrt{S_{\mu}} \sim N(0, 1)$; precise confidence interval construction with appropriate quantiles.

Question 4

- (a) Explain the rationale behind the likelihood ratio test and why the likelihood ratio test statistic takes values larger than one. Describe how this test may be used in the presence of nuisance parameters.

(8 marks)

Reading for this question

Sections 5.6 and 5.7 of the course present the relevant material on likelihood ratio tests and testing in the presence of nuisance parameters.

Approaching the question

The following is from the subject guide. A summary of the below is sufficient for full marks.

The likelihood ratio test statistic is connected with maximum likelihood estimation. Consider an observed sample $\mathbf{x}$. Hence the observed value of the likelihood ratio test statistic is:

$$LR(\mathbf{x}) = \frac{\sup_{\theta \in \Theta} L_{\mathbf{X}}(\theta; \mathbf{x})}{\sup_{\theta \in \Theta_0} L_{\mathbf{X}}(\theta; \mathbf{x})}.$$

The numerator is the *global maximum* of the likelihood function, attained at the maximum likelihood estimate $\hat{\theta}$. The denominator is the maximum of the likelihood function *under the constraint* that $\theta \in \Theta_0$. Let $\hat{\theta}_0$ denote the maximum value of this constrained likelihood function. Hence the observed value of the likelihood ratio test statistic can be expressed as:

$$LR(\mathbf{x}) = \frac{L_{\mathbf{X}}(\hat{\theta}; \mathbf{x})}{L_{\mathbf{X}}(\hat{\theta}_0; \mathbf{x})}.$$

This value must be ≥ 1 (since the constrained parameter estimate, $\hat{\theta}_0$, cannot be better than the unconstrained estimate, $\hat{\theta}$, in the sense that the associated likelihood value cannot be larger), although rejection of H_0 only occurs if the value is significantly greater than 1, i.e. when $\hat{\theta}_0$ is significantly worse than $\hat{\theta}$.

Now suppose that θ can be split into two groups of parameters: ψ being the main parameters of interest, and λ being the parameters which are of little interest – known as **nuisance parameters**.

Using this specification, we can think of H_0 as imposing a constraint on the model being fitted. The constraint is that ψ must take the values specified by ψ_0 . Hence the likelihood ratio test statistic compares two models, each with a relative advantage.

- The **unconstrained model** will provide a better fit since the associated likelihood will be larger. We denote the unconstrained maximum likelihood estimate by:

$$\hat{\theta} = (\hat{\psi}, \hat{\lambda}).$$

- The **constrained model** has fewer parameters requiring estimation. If the dimension of ψ is k , then the values of the k parameters contained in ψ are fixed, such that we only need to estimate the nuisance parameters which are in λ , where $\dim(\lambda) < \dim(\theta)$. Under the constraint that $\psi = \psi_0$, the maximum likelihood estimate of λ is generated by maximising:

$$L_{\mathbf{X}}(\psi_0, \lambda; \mathbf{x})$$

with respect to λ . We denote the resulting estimate as $\hat{\lambda}_0$. Hence the constrained maximum likelihood estimate of θ is:

$$\hat{\theta}_0 = (\psi_0, \hat{\lambda}_0).$$

When testing:

$$H_0 : \psi = \psi_0, \lambda \in \Lambda \quad \text{vs.} \quad H_1 : \psi \neq \psi_0, \lambda \in \Lambda$$

the likelihood ratio test statistic is:

$$LR(\mathbf{x}) = \frac{\sup_{\psi, \lambda} L_{\mathbf{X}}(\psi, \lambda; \mathbf{x})}{\sup_{\lambda} L_{\mathbf{X}}(\lambda; \psi_0, \mathbf{x})}.$$

Common errors: Confusing numerator/denominator; not explaining why $LR \geq 1$ (unconstrained likelihood cannot be smaller than constrained); vague treatment of nuisance parameters; failing to distinguish parameters of interest ψ from nuisance λ ; not explaining that λ is estimated under both H_0 and H_1 ; missing the model comparison interpretation.

Excellent answers provided clear intuition that LR compares the global maximum to the constrained maximum; explicit explanation why the unconstrained maximum likelihood estimator yields a higher likelihood; systematic description of nuisance parameter handling (fixing $\psi = \psi_0$ while optimising over λ); understanding the parsimony-fit trade-off; recognition that rejection requires LR significantly exceeding 1; complete mathematical specification of both optimisation problems.

- (b) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from the continuous uniform distribution on $[0, \theta]$, where $\theta > 0$ is unknown, for which the probability density function is:

$$f_X(x; \theta) = \begin{cases} \theta^{-1} & \text{for } 0 \leq x \leq \theta \\ 0 & \text{otherwise.} \end{cases}$$

- i. Find the most powerful test for testing $H_0 : \theta = 1$ against $H_1 : \theta = 2$ and such that it *never* rejects H_0 when H_0 is true.

(7 marks)

- ii. For $n = 4$, for the above test compute the power of the test *and* the probability of a Type II error.

(5 marks)

Reading for this question

Section 5.5 of the course presents most powerful tests.

Approaching the question

- i. Consider an observed sample $\mathbf{x} = \{x_1, x_2, \dots, x_n\}$ and let $x_{(n)} = \max_i x_i$ and $x_{(1)} = \min_i x_i$. Due to independence, the likelihood function is:

$$L_{\mathbf{X}}(\theta; \mathbf{x}) = \prod_{i=1}^n f_{X_i}(x_i; \theta) = \prod_{i=1}^n \theta^{-1} I(x_{(n)} \leq \theta) I(x_{(1)} \geq 0).$$

By the Neyman–Pearson lemma the most powerful test has a rejection region defined as:

$$\begin{aligned} R &= \{\mathbf{x} \text{ such that } L_{\mathbf{X}}(\theta = 2; \mathbf{x}) \geq L_{\mathbf{X}}(\theta = 1; \mathbf{x})k\} \\ &= \{\mathbf{x} \text{ such that } 2^{-n} I(x_{(n)} \leq 2) \geq I(x_{(n)} \leq 1)k\} \end{aligned}$$

where $k > 0$ and is such that $P(\mathbf{x} \in R | H_0) = 0$. This is equivalent to defining the rejection region:

$$R = \{\mathbf{x} \text{ such that } x_{(n)} > 1\}.$$

Indeed, we have that:

$$P(\mathbf{x} \in R | H_0) = P(X_{(n)} > 1 | \theta = 1) = 0.$$

- ii. The power of the test is:

$$\begin{aligned} P(\mathbf{x} \in R | H_1) &= P(X_{(n)} > 1 | \theta = 2) \\ &= 1 - P(X_{(n)} \leq 1 | \theta = 2) \\ &= 1 - (P(X_1 \leq 1 | \theta = 2))^n \\ &= 1 - \frac{1}{2^n} \end{aligned}$$

and for $n = 4$ the power is $15/16 = 0.9375$. Hence the probability of a Type II error is:

$$1 - P(\mathbf{x} \in R | H_1) = 1 - 0.9375 = 0.0625.$$

Common errors (i): Misapplying the Neyman–Pearson lemma; ignoring support constraints in likelihood indicators; not recognising that size = 0 requirement determines k uniquely; failing to simplify to the rejection region $x_{(n)} > 1$; incorrect handling of the likelihood ratio when one likelihood is zero. In part (ii): Not using the order statistic CDF formula; computational errors calculating $P(X \leq 1 | \theta = 2) = 1/2$; confusing power with Type II error; missing that $P(\text{Type II error}) = 1 - \text{power}$.

Excellent answers had the explicit likelihood with support indicators; systematic Neyman–Pearson application showing inequality simplification; clear justification why $x_{(n)} > 1$ achieves size zero; accurate power calculation using complement and order statistic theory; correct relationship between power and the probability of a Type II error; all numerical results clearly stated.